Orbit distance and strict support deficits

Human Compression project: bounded L044 proof component

Use the standing assumptions of the internal-orbit-block note: $G \leq A_n$, $n \geq 12$, has a transitive connected string C-marking with every maximal omission intransitive; a unique eligible interior i-edge joins $a \in A$ to $b \in B$; $L = G_{<i}$ fixes B pointwise and is intransitive on A. The L-orbits in A have a common size k, their actions are faithful, and $\beta = a^L$ is exchanged with a distinct β' by $\tau = \rho_{i+1}$. All later generators fix β pointwise, and those of index at least $i+3$ fix β' pointwise as well. These facts have already been proved completely in that note.

Lemma 1 (Distance to an active commuting orbit)**.** *If $h > i$ and ρ_h acts nontrivially on A, put $d = h - i$. For $d \geq 2$, at least d distinct L-orbits occur in A. If the first orbit reached on which ρ_h acts nontrivially is moved to a different orbit by ρ_h, at least $d+1$ distinct orbits occur. In all cases $k \geq i+1$.*

Proof. Consider the orbit graph of L using only generators with indices strictly between i and h, starting at β. Some reachable orbit has nontrivial ρ_h-action. Indeed, if ρ_h fixed all those orbits pointwise, every higher indexed generator would fix them pointwise too: it fixes β and commutes with all the generators used in the paths. Their union would be invariant under G_i. Transitivity of G_i on A would then make it all of A, contrary to the nontrivial ρ_h-action.

Take a shortest orbit path to such an orbit, of length t, and let w be its word in distinguished generators. Its orbits are distinct, and a^w is moved by ρ_h: a nontrivial permutation commuting with the transitive L-action on an invariant orbit has no fixed point; if it moves that orbit the assertion is immediate.

Every index j strictly between i and h occurs in w. Otherwise split the word into factors with indices less than j and greater than j. They commute. The latter factor fixes a, since all its indices are at least $i+2$. The former commutes with ρ_h, which also fixes a. Thus a^w would be fixed by ρ_h, a contradiction. Consequently $t \geq d-1$, proving the first count. All earlier orbits on the path are fixed pointwise by ρ_h, by the choice of the first active orbit. If the final orbit is moved, its image is not one of those earlier fixed orbits, giving the extra orbit claimed. Finally the faithful degree-k action of L retains its independent distinguished family of size i; the paid Whiston bound gives $i \leq k-1$. □

Proposition 2 (The branch with tail supported on the second orbit)**.** *Let $h > i$ be the largest index acting nontrivially on both A and B, and suppose $h < r-1$ and $G_{>h}$ fixes A pointwise. Put $X = \mathrm{supp}(G_{>h})$ and $F = n - |X|$. Then*

$$X \subseteq B \setminus \{b\}, \qquad F \geq 2h+1.$$

If $h \geq i+2$, the inequality is strict. If $F = 2h+1$, then $h = i+1$, $k = i+1$, $A = \beta \cup \beta'$, and the fixed set is exactly $A \cup \{b\}$. The faithful action of L on either orbit is the full S_k.

Proof. The accepted fixed-prefix lemma gives, in B, a simple path from b to a point moved by ρ_h, using only earlier indices and excluding i. Its $d = h - i$ or more vertices are fixed by $G_{>h}$. Together with all of A, this gives

$$F \geq |A| + d. \tag{1}$$

The support inclusion also follows from fixation of A and of the crossing endpoint b.

For $d = 1$, there are at least two L-orbits, so $F \geq 2k + 1 \geq 2i + 3 = 2h + 1$. Equality forces $k = i + 1$, exactly two orbits in A, and no fixed vertex outside $A \cup \{b\}$. Whiston's equality theorem makes the faithful L-action on a single orbit S_k.

For $d = 2$, the group generated by indices between i and h is just $\langle \tau \rangle$. The first active orbit is β'. If ρ_h moves it, Lemma 1 gives

$$F \geq 3k + 2 \geq 3i + 5 > 2i + 5 = 2h + 1.$$

If it preserves it, the complete internal-block note applies and gives the stronger $F \geq 2h + 3$.

For $d \geq 3$, Lemma 1 and (1) give

$$F \geq d(k + 1) \geq d(i + 2) = 2h + 1 + \big(i(d - 2) - 1\big) \geq 2h + 1.$$

Equality could occur only when $i = 1$, $d = 3$, $k = 2$, and A consists of exactly three L-orbits. But then $L = \langle \rho_0 \rangle$, and on each of those two-point orbits ρ_0 is a transposition. It fixes B pointwise, so it would be a product of exactly three transpositions, contrary to evenness. This excludes equality and completes the proof in every case. □

Exact scope and attribution. This is a new orbit-count proof of the numerical support part of L044 Proposition 5.23 when the final tail fixes the first orbit, retaining the source's commuting-block idea and all hypotheses. The source is Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, physical pages 32–33. Whole orbit counts replace the need to infer disjointness of transported point paths. The paid fixed-prefix, internal-block and Whiston inputs remain explicit. No finite computation, new formal declaration, whole-original or question credit is asserted. The branch where the final tail acts on A, the full coloured equality graph, and subsequent full-IP rigidity are not proved by this note.